

1.

Author (AuthorID, Name, Country)

AuthorID is the primary key

Book (BookID, Title, PubYear, AuthorID)

BookID is the primary key

AuthorID is a foreign key from Author table

Member (MemberID, Name, Email, JoinDate)

MemberID is the primary key

Borrow (BorrowID, BookID, MemberID, BorrowDate, ReturnDate)

BorrowID is the primary key

BookID is a foreign key from Book table

MemberID is a foreign key from Member table

2.

-- Author Table

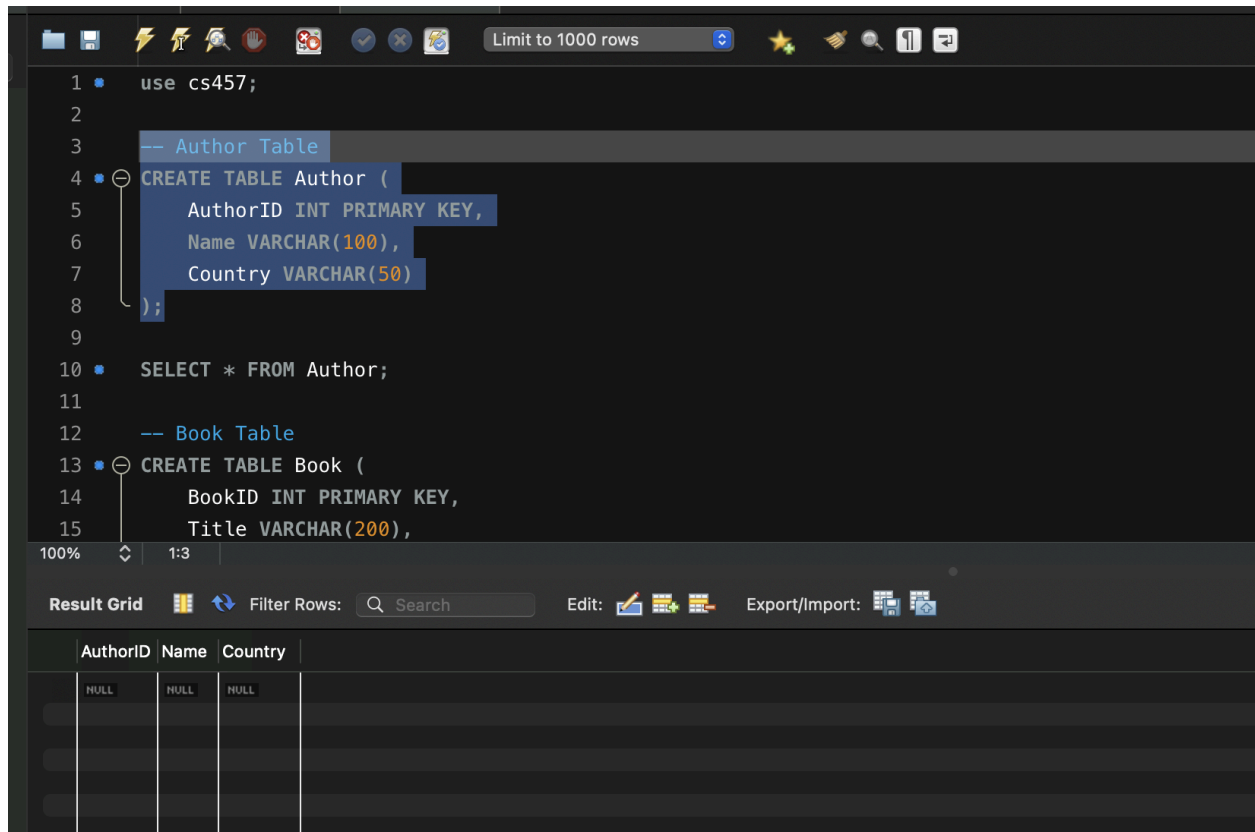
CREATE TABLE Author (

AuthorID INT PRIMARY KEY,

Name VARCHAR(100),

Country VARCHAR(50)

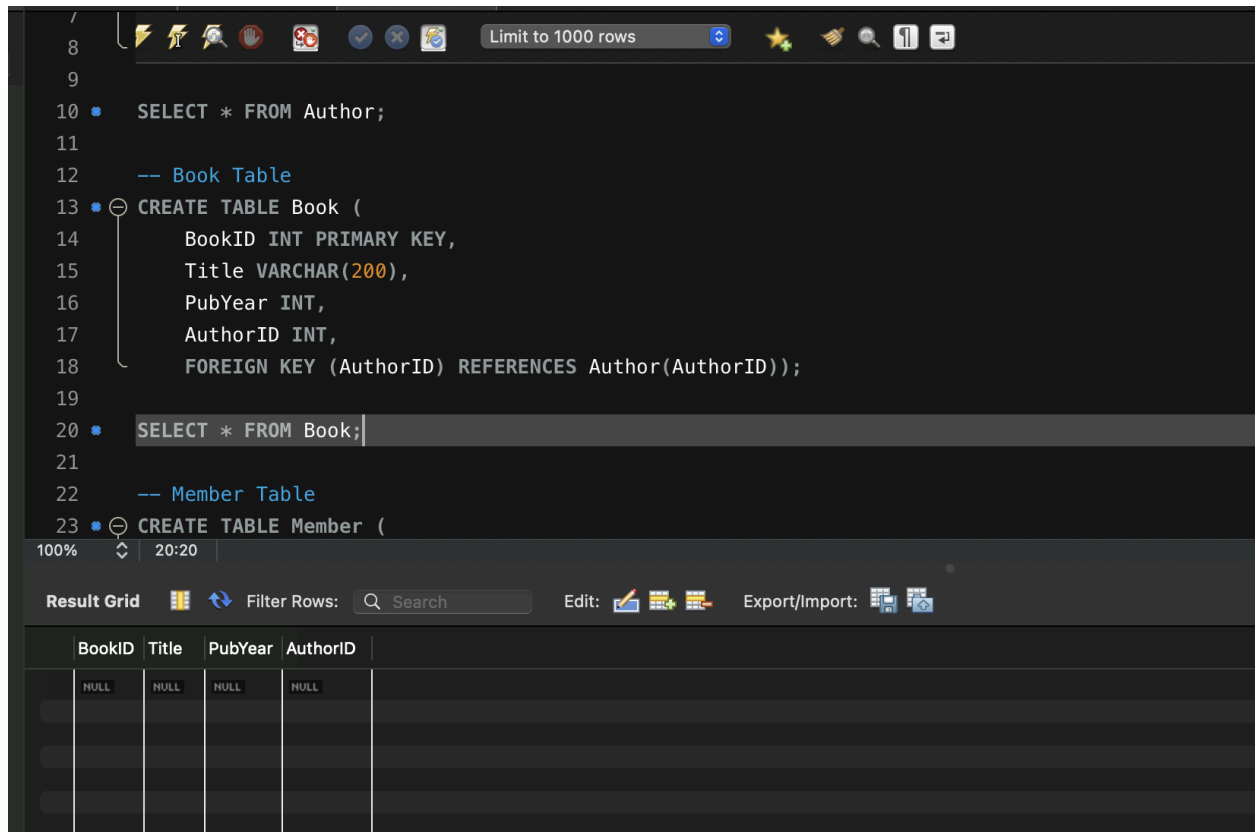
);



-- Book Table

```
CREATE TABLE Book (
    BookID INT PRIMARY KEY,
    Title VARCHAR(200),
    PubYear INT,
    AuthorID INT,
    FOREIGN KEY (AuthorID) REFERENCES Author(AuthorID));
```

```
SELECT * FROM Book;
```



-- Member Table

CREATE TABLE Member (

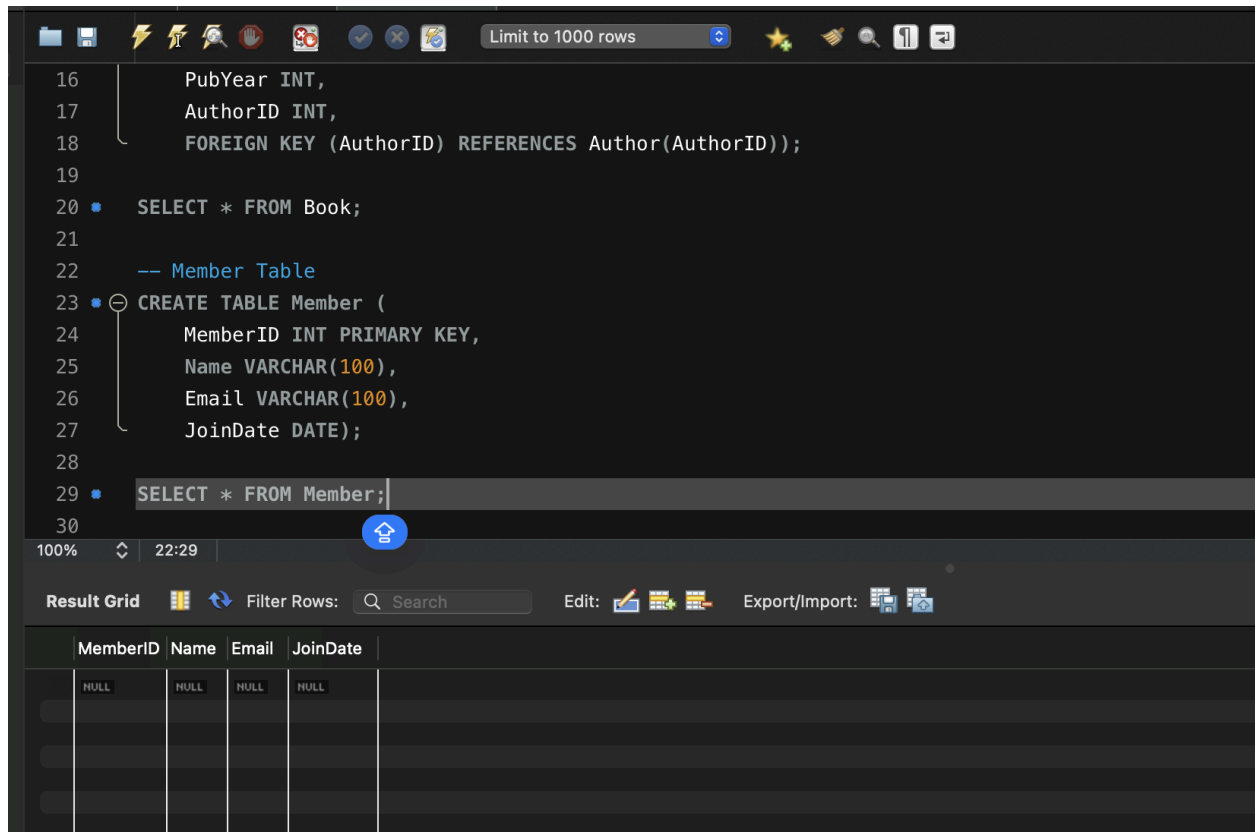
MemberID INT PRIMARY KEY,

Name VARCHAR(100),

Email VARCHAR(100),

JoinDate DATE);

SELECT * FROM Member;



-- Borrow Table

CREATE TABLE Borrow (

BorrowID INT PRIMARY KEY,

BookID INT,

MemberID INT,

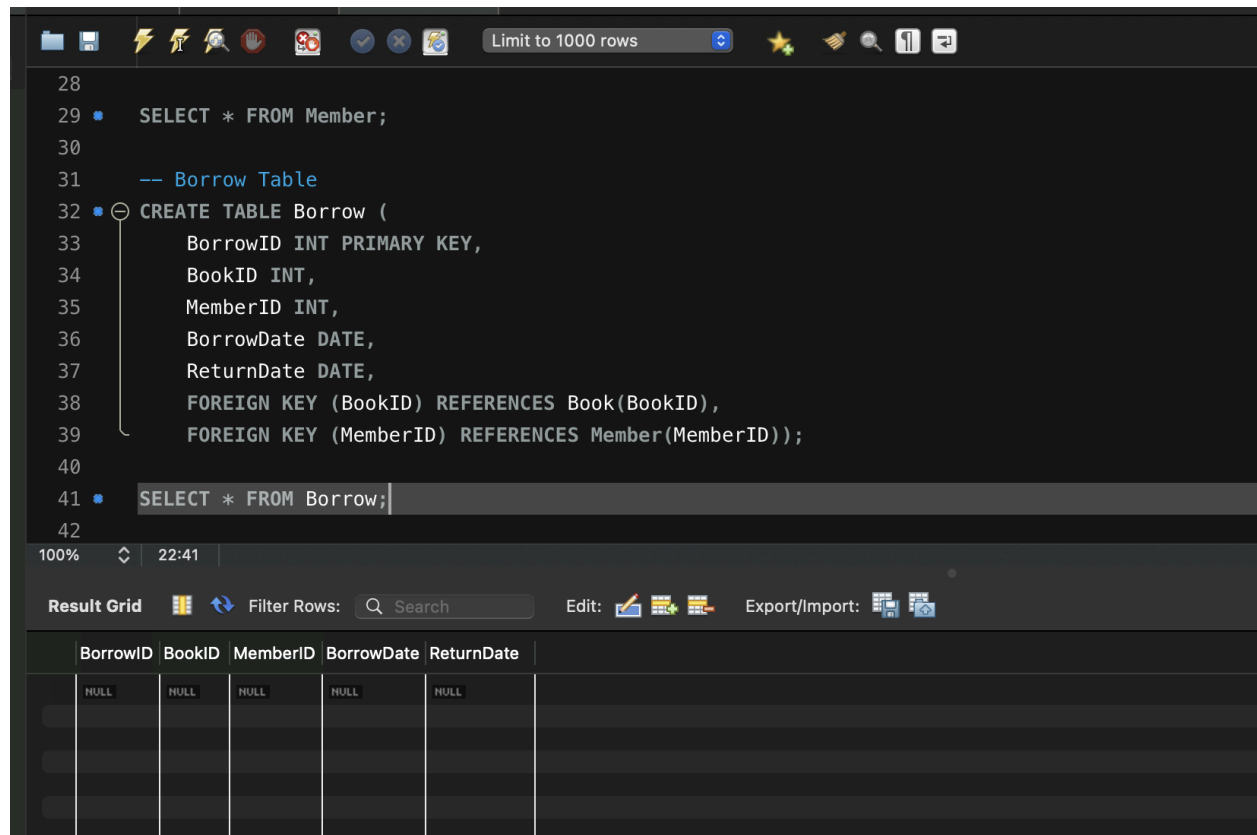
BorrowDate DATE,

ReturnDate DATE,

FOREIGN KEY (BookID) REFERENCES Book(BookID),

FOREIGN KEY (MemberID) REFERENCES Member(MemberID));

SELECT * FROM Borrow;



3.

```
-- Insert into Author
INSERT INTO Author (AuthorID, Name, Country) VALUES
(1, 'J.K. Rowling', 'UK'),
(2, 'George R.R. Martin', 'USA'),
(3, 'Haruki Murakami', 'Japan');

SELECT * FROM Author;
```

```

36      BorrowDate DATE,
37      ReturnDate DATE,
38      FOREIGN KEY (BookID) REFERENCES Book(BookID),
39      FOREIGN KEY (MemberID) REFERENCES Member(MemberID));
40
41      -- Insert into Author
42      INSERT INTO Author (AuthorID, Name, Country) VALUES
43      (1, 'J.K. Rowling', 'UK'),
44      (2, 'George R.R. Martin', 'USA'),
45      (3, 'Haruki Murakami', 'Japan');
46
47      SELECT * FROM Author;
48
49      -- Insert into Book
50      INSERT INTO Book (BookID, Title, PubYear, AuthorID) VALUES

```

100% 1:47

Result Grid Filter Rows: Search Edit: Export/Import:

AuthorID	Name	Country
1	J.K. Rowling	UK
2	George R.R. Martin	USA
3	Haruki Murakami	Japan

-- Insert into Book

```

INSERT INTO Book (BookID, Title, PubYear, AuthorID) VALUES
(101, 'Harry Potter and the Sorcerer's Stone', 1997, 1),
(102, 'Harry Potter and the Chamber of Secrets', 1998, 1),
(201, 'A Game of Thrones', 1996, 2),
(301, 'Kafka on the Shore', 2002, 3);

```

```

SELECT * FROM Book;

```

The screenshot shows a database management interface with a SQL editor and a result grid. The SQL editor contains the following queries:

```
45 (3, 'Haruki Murakami', 'Japan');
46
47 • SELECT * FROM Author;
48
49 -- Insert into Book
50 • INSERT INTO Book (BookID, Title, PubYear, AuthorID) VALUES
51 (101, 'Harry Potter and the Sorcerer's Stone', 1997, 1),
52 (102, 'Harry Potter and the Chamber of Secrets', 1998, 1),
53 (201, 'A Game of Thrones', 1996, 2),
54 (301, 'Kafka on the Shore', 2002, 3);
55
56 • SELECT * FROM Book;
57
58 -- Insert into Member
59 • INSERT INTO Member (MemberID, Name, Email, JoinDate) VALUES
```

The result grid displays the data for the 'Book' table:

BookID	Title	PubYear	AuthorID
101	Harry Potter and the Sorcerer's Stone	1997	1
102	Harry Potter and the Chamber of Secrets	1998	1
201	A Game of Thrones	1996	2
301	Kafka on the Shore	2002	3
NULL	NULL	NULL	NULL

```
-- Insert into Member
INSERT INTO Member (MemberID, Name, Email, JoinDate) VALUES
(1, 'Alice Johnson', 'alice@example.com', '2023-01-15'),
(2, 'Bob Smith', 'bob@example.com', '2024-03-10');

SELECT * FROM Member;
```

The screenshot shows a SQL IDE interface. The top toolbar includes icons for lightning bolt, lightning bolt with a circle, a hand, a red X, a checkmark, a blue X, and a document with a lightning bolt. A dropdown menu shows "Limit to 1000 rows". The script editor contains the following SQL code:

```
53
54
55
56 • SELECT * FROM Book;
57
58 -- Insert into Member
59 • INSERT INTO Member (MemberID, Name, Email, JoinDate) VALUES
60   (1, 'Alice Johnson', 'alice@example.com', '2023-01-15'),
61   (2, 'Bob Smith', 'bob@example.com', '2024-03-10');
62
63 • SELECT * FROM Member;
64
65 -- Insert into Borrow
66 • INSERT INTO Borrow (BorrowID, BookID, MemberID, BorrowDate, ReturnDate) VALUES
67   (1, 101, 1, '2024-05-01', '2024-05-20'),
68   (2, 201, 2, '2024-06-10', NULL);
69
```

Below the script editor, the "Result Grid" is displayed. It has a search bar and icons for "Filter Rows", "Edit", and "Export/Import". The grid shows the results of the last query:

MemberID	Name	Email	JoinDate
1	Alice Johnson	alice@example.com	2023-01-15
2	Bob Smith	bob@example.com	2024-03-10
NULL	NULL	NULL	NULL

-- Insert into Borrow

```
INSERT INTO Borrow (BorrowID, BookID, MemberID, BorrowDate, ReturnDate) VALUES
(1, 101, 1, '2024-05-01', '2024-05-20'),
(2, 201, 2, '2024-06-10', NULL);
```

```
SELECT * FROM Borrow;
```



```
58 -- Insert into Member
59 • INSERT INTO Member (MemberID, Name, Email, JoinDate) VALUES
60 (1, 'Alice Johnson', 'alice@example.com', '2023-01-15'),
61 (2, 'Bob Smith', 'bob@example.com', '2024-03-10');
62
63 • SELECT * FROM Member;
64
65 -- Insert into Borrow
66 • INSERT INTO Borrow (BorrowID, BookID, MemberID, BorrowDate, ReturnDate) VALUES
67 (1, 101, 1, '2024-05-01', '2024-05-20'),
68 (2, 201, 2, '2024-06-10', NULL);
69
70 • SELECT * FROM Borrow;
71
```

100% 1:70

Filter Rows:

Q Search

Export/Import:

[illegible]